

Infelicity with Homogeneity: Focus, not Presupposition*

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Abstract

Wehbe (2022) and Guerrini and Wehbe (2024) present new arguments that homogeneity with definite plurals is the result of a homogeneity presupposition. Their arguments are based on the observation that definite plurals are infelicitous in certain contexts where one expects them to be infelicitous if homogeneity is a presupposition. We argue that a closer scrutiny of the empirical picture reveals that infelicity in most of the cases they discuss is due to independent requirements from focus, and suggests the opposite conclusion, namely that there is no homogeneity presupposition. We also point out that other infelicitous examples which cannot be explained based on focus are explained if homogeneity effects are due to implicature computation.

Keywords: homogeneity · presupposition · focus · implicature · definite plurals · questions

1 Introduction

The phenomenon of homogeneity in definite plurals can be illustrated in (1). A basic question is how to explain the (near-)universal force of definite plurals in upward monotone environments as in (1a) and its (near-)existential force in downward monotone environments as in (1b) (Löbner 1987, 2000; Schwarzschild 1994; Krifka 1996; Gajewski 2005; Magri 2014; Križ 2015, 2016; Križ and Spector 2017; Bar-Lev 2021; Feinmann 2021; Chatain 2022, a.o.).

- (1) a. Ann read the books. $\approx$ Ann read **all** of the books
 b. Ann didn't read the books. $\approx$ Ann read **none** of the books

Some authors have claimed that homogeneity effects in definite plurals are the result of a homogeneity presupposition (HP), as in (2) (Schwarzschild 1994; Löbner 2000; Gajewski 2005, a.o.)

- (2) **Homogeneity presupposition (HP):**
 Ann read all of the books or none of them (**All** $\vee$ **None**)

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The HP theory is much debated, mainly surrounding questions of projection (Spector 2013; Križ 2015, 2016; Križ and Spector 2017; Fox 2018). Recently, a new argument for HP was put forth by Wehbe (2022); Guerrini and Wehbe (2024) (henceforth WG) based on a property of presupposition accommodation. Specifically, WG argue that assuming HP for definite plurals explains what we will call ‘infelicity with homogeneity’, illustrated by the infelicity of (3a) compared to the felicity of (3b), given that presupposition accommodation should not make the assertive component of meaning trivial; we will get into the details of their argument in the next section. (Throughout, capital letters represent focal stress. For now we will set aside the fact that stress on the quantifier is required in (3b) and other examples we discuss below; this will play an important role in our proposal and we will return to it in section 4.)

- (3) I knew that Ann read at least some of the books. But today I learned that...
- a. #she read the books.
 - b. she read ALL the books.

In this note, we argue that infelicity with homogeneity as in (3a), which WG used to motivate HP, has an alternative explanation, which can explain infelicity even in cases where assuming HP does not help. Specifically, our alternative account relies on independently needed constraints on focus. Our alternative thus undermines WG’s arguments in support of assuming HP. We further argue that assuming HP is not only unnecessary for explaining infelicity in examples like (3a), but rather it predicts infelicity in cases where there is in fact no infelicity, and is thus more harmful than helpful.

The note is structured as follows. In section 2, we review the argument by WG for HP. In section 3 we argue that WG’s HP-based account of infelicity with homogeneity is both too weak (failing to predict infelicity in some cases) and too strong (predicting infelicity for some felicitous sentences). Section 4 then points out that there is a connection between infelicity with homogeneity and focus, motivating the alternative account we present in section 5 according to which infelicity with homogeneity is due to conditions on focus. Section 6 is concerned with one type of infelicity with homogeneity that is explained by WG’s account but not by our focus-based account, and points out that the problem disappears if one adopts the implicature account of homogeneity (Magri 2014; Bar-Lev 2018, a.o.). Section 7 concludes and discusses a possible direction for reconciling the HP view with our observations, as well as infelicity with homogeneity in various domains where homogeneity effects have been observed.

2 The argument from PAI

Post-Accommodation Informativity (PAI) is a general principle that constrains sentences with presupposition to only be felicitous if the assertive component of the sentence is informative after accommodating the presupposition (Heim 2015; Doron and Wehbe 2022).

- (4) **Post-Accommodation Informativity (PAI):**

A sentence S_p (presupposing p) can be uttered felicitously only if S_p is informative w.r.t. the common ground and QUD after presupposition accommodation.

PAI explains why a case like (5a), involving the presupposition trigger *stop*, is infelicitous: as (6) shows, given the context-setting sentence and after the presupposition of (5a) is accommodated, there is no more

room for the assertion encoded by (5a) to be informative.

- (5) I knew that Mary doesn't smoke now. But today I learned that. . .
- a. #she stopped smoking.
 - b. she used to smoke.

- (6) In (5a): $\underbrace{\neg(\text{Mary smokes now})}_{\text{Context}} \wedge \underbrace{\text{Mary used to smoke}}_{\text{Presupposition}}$
 $\Rightarrow$ Mary used to smoke and doesn't smoke now

As previewed, Wehbe and Guerrini and Wehbe use PAI to furnish a new argument for a homogeneity presupposition for definite plurals. The paradigm in (7) is repeated from (3). The infelicity with homogeneity in (7a) is explained by PAI on the assumption that sentences with definite plurals trigger homogeneity as a presupposition. As (8) schematically illustrates, given the contextual setup which excludes worlds in which Ann read none of the books, (7a) becomes uninformative after accommodating the HP, explaining its oddness. By contrast, in (7b) there is no HP and consequently PAI isn't violated.

- (7) I knew that Ann read at least some of the books. But today I learned that...
- a. #she read the books.
 - b. she read ALL the books.

- (8) In (7a): $\underbrace{\neg\text{None}}_{\text{Context}} \wedge \underbrace{(\text{All} \vee \text{None})}_{\text{HP}} \Rightarrow \text{All}$

A parallel reasoning applies to the paradigm in (9) in which the sentences are embedded under negation. HP+PAI for (9a) explains its infelicity, since the context-setting sentence contradicts **All**.

- (9) I knew that Ann didn't read all of the books. But today I learned that...
- a. #she didn't read the books.
 - b. she didn't read ANY of the books.

- (10) In (9a): $\underbrace{\neg\text{All}}_{\text{Context}} \wedge \underbrace{(\text{All} \vee \text{None})}_{\text{HP}} \Rightarrow \text{None}$

3 Problems for HP+PAI

3.1 HP+PAI is too weak

A further welcome prediction of HP+PAI is that it correctly predicts sentences with definite plurals to be bad as answers to questions that arguably carry an existence presupposition:¹

- (11) a. Did Ann read some of the books, or all of them?
(Existence presupposition: Ann read some of the books)
- b. #Ann read the books.

¹ We thank Danny Fox (p.c.) for pointing out to us the infelicity of sentences with definite plurals as answers to certain questions.

This is because, as (12) shows, the response in (11b) is uninformative after accommodation of HP together with the existence presupposition introduced by the alternative question in (11a).

$$(12) \quad \text{In (11b):} \quad \underbrace{\neg \text{None}}_{\text{Existence presupposition}} \wedge \underbrace{(\text{All} \vee \text{None})}_{\text{HP}} \Rightarrow \text{All}$$

However, we point out that HP makes the wrong prediction for questions with no existence presupposition, where the facts look much the same:²

- (13) a. (i) Did Ann read none of the books, some of the books, or all of the books?
(ii) Did Ann read none of the books, or all of them?
b. #Ann read the books.

HP+PAI doesn't explain (13), because the absence of the existence presupposition in the question does make the assertive component of the response informative, as schematized in (14).

$$(14) \quad \text{In (13b):} \quad \neg \text{None} \wedge \underbrace{(\text{All} \vee \text{None})}_{\text{HP}} \Rightarrow \text{All}$$

Another example that shows that HP+PAI is too weak is the following (due to Luka Crnić, p.c.):

- (15) #Yesterday she read some of the newspapers. Today she read the newspapers.

Note that the first sentence in (15) does not entail anything about what happened today, and specifically it does not entail that she read some of the newspapers today; but the second sentence is still infelicitous.

There is thus reason to cast doubt on the generality of WG's HP+PAI account of infelicity with homogeneity in section 2.

3.2 HP+PAI is too strong

Another problem which looks even more concerning for HP+PAI is that HP+PAI over-predicts infelicity for certain felicitous utterances. Consider (16), involving polarity(/'verum') focus.³

- (16) A: I know that Mary read some of the books, but I suspect that she didn't read all of them.
B: No, don't worry, she DID read the books.

The felicity of (16) is unexpected if we assume HP+PAI, because given A's context-setting sentence, B's assertion should be uninformative. Importantly, run-of-the-mill cases of PAI violations with uncontested presuppositions do not behave like this. (17B) illustrates with a PAI violation stemming from the presupposition of *stop*:⁴

² As Jad Wehbe (p.c.) points out, one can consider a stronger version of PAI that rules out a sentence as an answer to a question if the sentence has a presupposition that eliminates cells in the question; this would explain why (13b) is bad as an answer to (13a-i) if it has HP. However, it would not explain why (13b) is bad as an answer to (13a-ii) as well, where the question already presupposes HP. Furthermore, it would not seem to help explaining the infelicity of (15).

³ Adding a numeral in the definite plural (*the five books* rather than *the books*) does not affect the judgment for (16), and will guard against a possible explanation of its felicity based on an analysis of *the books* that involves domain restriction to the books that the speaker doesn't know whether Mary read.

⁴ Focus is sometimes able to rescue from PAI violations, as the felicity of (i) in the context of (17) demonstrates. We will get back to this point and its relevance to the HP theory in section 7.

- (17) A: I know that Mary doesn't smoke now, but I'm worried that she also didn't smoke in the past (in which case she is not a good fit for our experiment).
 B: #No, don't worry, she DID stop smoking.
 B': No, don't worry, she DID smoke in the past.

Given that PAI seems to hold quite generally, the felicity of (16B) then poses a problem for the assumption that HP exists. In section 5 we will discuss another example that poses a similar problem to HP (ex. (33)).

3.3 Intermediate summary

To take intermediate stock, HP for definite plurals can explain certain cases of infelicity with homogeneity, but it appears to be both too-weak (section 3.1) and too-strong (section 3.2). In the next two sections we will propose an alternative explanation of WG's facts. We will point out that independent conditions on focus placement seem to underlie the behavior of the felicitous and the infelicitous examples we've seen so far. This will lead us to present a focus-based analysis of the facts which does not assume HP.

4 Towards an alternative account: the homogeneity-focus generalization

As a first step towards an analysis, we want to point out that there is a close connection between infelicity with homogeneity in the examples we discussed so far and the stress pattern observed when the intended meanings are conveyed by overt quantifiers. As it turns out, in all the cases we have seen, when there is an overt quantifier, it must be stressed for the sentence to be felicitous. That is, the following generalization seems to hold:

(18) **The homogeneity-focus generalization:**

A sentence with a definite plural in a context *C* is infelicitous if its overt-quantificational counterpart requires stress on the quantifier in order to be felicitous in *C*.

Let us show that the generalization in (18) captures all the cases of infelicity with definite plurals that we have seen so far. The infelicity in (19a), (20a), (21b-i) and (22a) correlates with the fact that the quantifier in (19b), (20b), (21b-ii) and (22b) must be stressed for the sentence to be felicitous (observe that for the definite plural version of the examples, by contrast, there is no stress pattern that makes them felicitous; in particular, stress on the definite article doesn't save the sentences).

- (19) I knew that Ann read at least one of the books. But today I learned that...
 a. #she read the books. [= (3a)/(7a)]
 b. she read {ALL, #all} the books.
- (20) I knew that Ann didn't read at least one of the books. But today I learned that...
 a. #she didn't read the books. [= (9a)]
 b. she didn't read {ANY, #any} of the books.
- (21) a. (i) Did Ann read some of the books, or all of them?
 (i) No, don't worry, she STOPPED smoking.

- (ii) Did Ann read none of the books, some of the books, or all of them?
- (iii) Did Ann read none of the books, or all of them?
- b. (i) #Ann read the books. [= (11b)/(13b)]
- (ii) Ann read {ALL, #all} the books.
- (22) a. #Yesterday she read some of the newspapers. Today she read the newspapers. [= (15)]
- b. Yesterday she read some of the newspapers. Today she read {ALL, #all} the newspapers.

The homogeneity-focus generalization also implies that if a sentence with a definite plural is felicitous, then its overt-quantificational counterpart does not require stress on the quantifier. This is indeed borne out when we consider (16), as (23) demonstrates:

- (23) A: I know that Mary read some of the books, but I suspect that she didn't read all of them.
- B: No, don't worry, she DID read the books. [= (16)]
- B': No, don't worry, she DID read all the books.

5 Proposal

5.1 Some common assumptions on focus

The homogeneity-focus generalization suggests that the infelicity with homogeneity in all the cases we discussed has to do with considerations of focus. As a clue to why that should be, we may first ask why overt quantifiers require stress in (19)–(23). The answer looks rather obvious and relates to the known connection between focus and Questions Under Discussion (QUD; Roberts 1996a): The contexts in all of these examples make it so that the most natural QUD cares about the distinction between quantifiers (*none/some/all*); we will call this type of question a 'quantificational question'. This is immediately visible when such a question is made explicit (as in (21)); in this case standard assumptions about focus require focus on the quantifier, as in (24b).

- (24) a. Did Ann read none of the books, some of the books, or all of them?
- b. Ann read [all]_F of the books.

Specifically, the requirements on focus placement that we adopt are in (25). They make it so that (24b) is the only possible F-structure of the sentence: *all* must be dominated by an F-marked constituent given a *Q* like (24a) in order to satisfy the congruence requirement in (25a) (see Rooth 1992, Roberts 1996b, Büring 2019, Katzir 2024, a.o.), and F-marking on any constituent other than *all* would violate the requirement on focus minimization in (25b) (Schwarzschild 1999):

- (25) **Focus Placement:**
 - a. **Question-answer congruence:** The focus structure of a sentence *S* must be congruent to an explicitly-provided or implicitly-accommodated question *Q* that *S* answers, such that $Q \subseteq \llbracket S \rrbracket_f$.
 - b. **Focus minimization:** F-mark as little as is necessary to satisfy question-answer congruence.

We assume that *some* and *none* are focus alternatives of *all*.⁵ The focus value of (24b) is then as in (26), which is congruent to the question in (24a).

- (26) $\llbracket (24b) \rrbracket_f = \{ \text{Ann read none of the books, Ann read some of the books,} \\ \text{Ann read all of the books} \}$

Furthermore, given the following constraint on the realization of focus, *all* would have to be stressed given the structure in (24b) (see Truckenbrodt 1995; Büring 2016, a.o.):

- (27) **Focus Realization:**
The highest stress in a sentence (more generally, in a focus domain) lies on an F-marked phrase in the sentence. (adapted from Büring 2016:164)

5.2 Explaining infelicity with homogeneity based on focus

We are then left with the question why we get infelicity with homogeneity in such contexts:

- (28) a. Did Ann read none of the books, some of the books, or all of them?
b. #Ann read the books.

A natural explanation for the infelicity of (28b) suggests itself: the focus requirements cannot be met due to the lack of an overt quantifier. Our proposal is based on this intuition: in a nutshell, the lack of an overt quantifier against an (underlying) quantificational QUD means that either F-marking does not satisfy the F-placement requirements in (25) to begin with, or, even if it does, it will fail to be realized phonologically, violating the constraint on F-realization in (27).

We hold that this explanation extends to all the infelicitous cases with definite plurals we've seen so far, with or without an overt question, cf. our discussion in section 4 (examples (19a), (20a), (21b-i) and (22a)). In all of them, the discourse is structured in a way that leads us to accommodate the offending quantificational question, even where no such question is overtly provided (as also corroborated by the obligatoriness of stress on the quantifiers in the overt quantificational counterparts). Take (19a) for example. Its infelicity stems from the fact that the preceding discourse in (19) supports no QUD for which F-placement and F-realization can both be met. Instead it leads one to accommodate a quantificational QUD: $\{ \text{Ann read some of the books, Ann read all of the books} \}$.

This line of explanation is consistent with all theories of how homogeneity actually arises. In all of them, there is nothing overt in the structure of (28b) that contributes quantification. On some accounts the quantificational force of the sentence comes from extra-syntactic sources (e.g., Križ 2015; Križ and Spector 2017), which means that there is no way to F-mark anything in the structure so that F-placement would be satisfied if *Q* is a quantificational question. On other accounts there is a covert operator in the structure that introduces quantification (e.g., Gajewski 2005; Bar-Lev 2021; Guerrini and Wehbe 2024), which under certain assumptions about its focus alternatives might make it so that F-placement can be satisfied, but even in that case F-realization won't be satisfiable due to the covert nature of the operator.

⁵ There are reasons to assume that *none* is not a formal focus alternative of *all*, given arguments that it is composed of negation and an existential quantifier (Zeijlstra 2011; Penka 2011, a.o.) and under the assumption that focus alternatives are no more complex than the prejacent (Katzir 2007; Fox and Katzir 2011). But even on these accounts it should be an alternative given the question in (24).

Considerations of focus then explain infelicity with homogeneity in all examples we have seen so far in a way that is not dependent on any specific theory of homogeneity. This undermines WG’s argument in favor of HP, though it is in principle compatible with the existence of HP.

5.3 Explaining felicity with homogeneity based on focus

Recall however that (16), repeated in (29), is felicitous despite HP+PAI predicting that it should be infelicitous, thus providing an argument against HP.

- (29) A: I know that Mary read some of the books, but I suspect that she didn’t read all of them.
 B: No, don’t worry, she DID read the books.

Our focus-based proposal, in contrast, accounts for the felicity in (29) (but that, of course, calls for an account of homogeneity that does not posit HP, otherwise it would run afoul of PAI). Let us see how. We have already seen in section 4 that, in line with the homogeneity-focus generalization, the context in (29) does not require stress on the quantifier when it’s overt (ex. (23)):

- (30) B’: No, don’t worry, she DID read all the books.

This is because the context naturally supports the polar (non-quantificational) question in (31); this question is also what one would expect the focus value of (30) to be, given what’s usually assumed for polarity focus (e.g. Laka 1990; Wilder 2013; Roelofsen and Farkas 2015; Goodhue 2022).

- (31) {*She read all the books, She didn’t read all the books*}

This explanation for the possibility of unstressed *all* in (30) also explains the felicity of its definite plural counterpart in (29): Given homogeneity, this sentence is denotationally equivalent to the first member of (31), so one can derive for (29) a congruent set of alternatives with polarity focus.⁶

On our proposal, the felicity of (29), as opposed to the previous cases with definite plurals, stems from the fact that the salient QUD in this context plus the rules of focus realization conspire to allow it. Specifically, the polar QUD calls for polarity focus, and polarity focus is independently realized (in English) with stress on the auxiliary. This configuration then allows the definite plural to exhibit its homogeneous behavior, i.e. its denotational equivalence with *all* in UE environments, without causing a crash with the grammar of focus: Congruence is satisfied for (29) against the QUD in (31) just as much as it is satisfied for (30), and stress predictably falls on the auxiliary (in both (29) and (30)).

The point is general and does not affect just polarity focus. Whenever focus calculus independently predicts that main stress should fall on some other overt element in the sentence, our proposal predicts that a definite plural will be fine in a PAI-like context, whereas WG’s HP would still predict infelicity. Let us now demonstrate that our account makes the right prediction for cases other than polarity focus, using a slightly more involved kind of example and with a richer context. As a baseline, consider first

⁶ For the second member of (31) to be denotationally equivalent to a member of the focus value of the sentence, the negative alternative introduced by polarity focus would have to be non-homogeneous. On the implicature account of homogeneity we will adopt in section 6, this is what one would expect if the exhaustivity operator that derives the universal interpretation takes scope below the polarity head. Note that on a view where there is HP, deriving (31) as the set of alternatives for (29)B would require locally accommodating the homogeneity presupposition of (29)B below the polarity head. We will discuss the implications of having local accommodation in section 7.

(32a), which is odd no matter the prosody when uttered in the context of (32).

(32) John really likes to write phonology papers. Every day, he either works on some of them or all of them. He is in a cheerful mood now, so apparently. . .

- a. #he worked on his phonology papers today.
- b. he worked on {ALL, #all} of his phonology papers today.

(32a) is essentially a replication of WG's example in (3a). Just like in (3a), HP+PAI predicts infelicity, since the context entails that he worked on some of his phonology papers today; and just like in (3a), the most natural question in (32) is a quantificational question, as corroborated by the stress pattern in (33b), so one can also explain the infelicity of (33a) based on considerations of focus. Now consider the more elaborate context in (33), where the same sentence looks fine if stress falls on *phonology*, as in (33a).

(33) John really likes to write phonology papers. He also writes semantics papers, but he hates that. Every day, he either works on all of his phonology papers and none of his semantics papers, or all of his semantics papers and some of his phonology papers. He is in a cheerful mood, so apparently. . .

- a. he worked on his PHONOLOGY papers today.
- b. he worked on all of his PHONOLOGY papers today.

For HP+PAI, there should be no difference between (32) and (33): In both of them the context entails that John worked on some of his phonology papers today, leading to a violation of PAI. The fact that (33) is felicitous is then surprising for HP+PAI, and, like (29), suggests that HP does not exist.⁷ On our focus-based alternative, on the other hand, the felicity of (33a) is explained assuming that the most natural QUD in this context is as follows, as indicated by the stress pattern in (33b):

- (34) {*He worked on all of his phonology papers today*,
He worked on all of his semantics papers today}

Given this, the structure of the explanation carries over from the earlier case with polarity focus. (33a) and (33b) are denotationally equivalent given homogeneity; and (33a) is just as good as (33b) because, as opposed to their occurrence in the context in (32), the contextually-appropriate QUD is the one in (34), and the focus realization rules make it so that all conditions on F-marking are met, as now F-marking legitimately falls on overt material ('*phonology*').⁸

To sum up, all examples of infelicity with homogeneity so far are explained given some common assumptions about focus; these assumptions explain not only the infelicity of some examples that are explained by HP+PAI ((19a) and (20a)), but also the infelicity of examples that are not explained by HP+PAI ((21b-i) and (22a)) and the felicity of some examples that HP+PAI wrongly predicts to be

⁷ One may suspect that the inference from (33a) that he worked on none of his semantics papers today is enough to make the sentence non-trivial and thus avoid violating PAI. Note however that the context is set up in such a way that it is guaranteed that if he worked on all of his phonology papers than he worked on none of his semantics papers. So once HP (**All phonology** $\vee$ **No phonology**) is combined with the contextual assumptions (**Some phonology** $\wedge$ (**All phonology** $\rightarrow$ **No semantics**)), it follows that **All phonology** $\wedge$ **No semantics**, leaving no room for the sentence to be informative.

⁸ For some speakers the judgment regarding the felicity of (33a) might be subtle. We submit nonetheless that there is a contrast between the occurrence of this sentence in (33) vs. its occurrence in (32). The subtlety may be attributed to the fact that the context in (33) is intricate and that accommodating the right QUD from the sentence is somewhat of a complicated task.

infelicitous ((29) and (33a)). As we pointed out, the focus-based explanation does not require theoretical commitment to any particular theory of homogeneity, though if one would like to enjoy its benefits in accounting for the felicity of (29) and (33a), one would presumably want to stay away from accounts that posit HP (though see section 7 for a possible theoretical direction that a proponent of HP might pursue).

However, we will not be able to stay agnostic about the proper treatment of homogeneity for much longer. In the next section we will discuss an example of infelicity with homogeneity that WG bring up and that is not explained by our account so far. We will argue however that this case of infelicity is expected if one adopts the implicature account of homogeneity and consider some known properties of implicatures.

6 Triviality-based infelicity (but still without a presupposition)

Consider (35a) (due to Roger Schwarzschild, p.c.), which WG bring up as a further argument for HP+PAI.

(35) *Context: A monogamous society.*

- a. #John is not married to his teammates.
- b. John is not married to any of his teammates.

The overt-quantificational counterpart in (35b) does not require stress on the quantifier, which indicates that the context does not call for a quantificational question (i.e., a strange question like *is John married to none of his teammates, to some of his teammates, or to all of them?*), as one may expect given that the context is impoverished. Our focus-based account then does not account for the infelicity of (35a). In contrast, HP+PAI correctly predicts the infelicity of (35a), since the context entails that John is not married to all his teammates (a monogamous society), and, just like in (9), this means that the sentence violates PAI: the conjunction of $\neg\mathbf{All}$ (context) together with the $\mathbf{All} \vee \mathbf{None}$ presupposition (HP) makes the sentence (**None**) trivial.

We may seem to face a dilemma: Either we assume HP+PAI together with our focus story and explain everything we have seen so far except (29) and (33a), or we reject HP but then remain with no explanation for (35). Our goal in this section is to show that the infelicity of (35a) has an independent explanation that makes this dilemma disappear. Specifically, we suggest that the problem with (35a) lies in the fact that, on the implicature account of homogeneity (Magri 2014; Bar-Lev 2021), it has a reading in which it is contextually trivial, and, following work by Magri (2009a, 2011), this is enough to make a definite plural unacceptable in such a context.

On the implicature account of homogeneity, homogeneity effects arise due to lack of alternatives: The basic meaning of (36) is existential, as in (36a), which explains its existential meaning when embedded under negation and other DE contexts (recall our discussion in (1)). When unembedded, and more generally in UE contexts, its strengthened meaning is as in (36b) due to the fact that (36b) is not an alternative to (36).

(36) Ann read the books

- a. **Basic meaning** of (36): Ann read some of the books
- b. **Strengthened meaning** of (36): Ann read all of the books

On this view, we could explain the infelicity of (35a) if what's embedded under negation is strengthened: (37b) is a contextual contradiction, and if this strengthened meaning is embedded under negation in (35a), then the sentence ends up contextually tautologous.

(37) John is married to his teammates

- a. **Basic meaning** of (37): John is married to some of his teammates
- b. **Strengthened meaning** of (37): John is married to all of his teammates

Indeed, Magri (2009a,b, 2011) observed that this is how implicatures behave in general: Magri (2009a,b) points out that when strengthening a sentence leads to a contextual contradiction, the sentence is infelicitous, as in (38a); and Magri (2011) further observed that this behavior extends to embedding under negation (and other DE contexts), as in (38b):

(38) *Context: We know that Mary and John are compatriots.*

- a. #John or Mary come from a beautiful country.
- b. #It's not the case that John or Mary come from a beautiful country.

Without getting into too much detail, on Magri's account this happens due to the application of an exhaustivity operator $\mathcal{E}xh$ that derives strengthened meanings at every scope site, including under negation, together with constraints on its domain that force computation of the implicature when the derived meaning is contextually equivalent ($\Leftrightarrow_C$) to a contradiction, whether it is embedded under negation or not:

(39) $[\neg] \mathcal{E}xh$ (John or Mary come from a beautiful country)
 $\Leftrightarrow [\neg]$ (John or Mary but not both come from a beautiful country)
 $\Leftrightarrow_C [\neg] \perp$

Combining the theory of mandatory implicatures with the implicature account of homogeneity, one would then expect (35a) to be infelicitous for parallel reasons:⁹

(40) [Not] $\mathcal{E}xh$ (John is married to his teammates)
 $\Leftrightarrow [\neg]$ John is married to all his teammates
 $\Leftrightarrow_C [\neg] \perp$

The infelicity of (35a) then does not force us to assume HP; it is entirely expected if homogeneity is an

⁹ The way in which (40), and therefore infelicity, is predicted to arise, depends on the specifics of the accounts of mandatory implicatures and homogeneity-as-implicature. For reasons of space we do not go into the specifics; suffice it to say that (40) arises either on Magri's (2011) theory of mandatory implicatures when combined with Magri's (2014)'s account of homogeneity, or on Bar-Lev's (2023) take on mandatory implicatures when combined with either Magri's (2014) or Bar-Lev's (2021) account of homogeneity.

implicature instead.^{10,11}

7 Concluding remarks and open issues

To summarize, we proposed that infelicity with homogeneity is not due to HP as WG argued, but rather due to conditions on focus. We argued that a focus-based explanation has advantages over the HP+PAI view: (i) it can explain infelicity in some cases where HP+PAI is silent ((21b-i) and (22a)), and (ii) it can explain felicity in some cases where HP+PAI predicts infelicity ((29) and (33a)). We further pointed out that the only case WG present that is not explained by focus considerations (Schwarzschild's example in (35a)) is in fact expected if one adopts the implicature account of homogeneity. In the rest of this section we will discuss two open issues, one concerning a possible direction for reconciling the HP view with our arguments against it from (29) and (33a), and the other concerning (in)felicity with homogeneity in other domains.

7.1 Local Accommodation to the rescue?

Is there a way to shield the HP theory against our arguments from (29) and (33a)? One way to avoid PAI violations in these examples on a view that assumes HP could be with a brute-force LF operation responsible for relocating presuppositional information into the assertive component. Such a process is sometimes referred to as Local Accommodation (LA) (Heim 1983, Beaver and Krahmer 2001). If, for some reason, LA is possible in (29) and (33a), this could explain why PAI isn't violated in these examples despite there being HP: Once LA applied, the sentence would no longer carry HP and therefore there would be no PAI violation. Of course, one would have to say something about the distribution of LA, since if LA was always possible we would never expect PAI effects to arise (nor would we ever be able to detect presuppositions as such); the challenge would then be to explain what makes LA possible in cases like (29) and (33a).

One may look for clues in the non-default stress pattern in these examples. Indeed, Bassi et al. (2021) observed that LA (when computed under negation) must be accompanied by a particular stress pattern, and as we pointed out in footnote 4, certain stress patterns are able to rescue from PAI violations with (standard) presuppositions, which could be explained if stress is a condition on the application of LA. (41)-(43) illustrate this further with the triggers *stop*, singular *the* (uniqueness presupposition), and *again*, respectively.¹²

¹⁰ Spector (2025) recently proposed a different account of Magri effects, according to which a sentence is odd once it has a reading that is either tautological or contradictory. If (35a) has a tautologous reading along the lines of (40) (=John is not married to all his teammates), then, one would expect it to be infelicitous also on Spector's (2025) account, without needing to specifically assume an implicature account of homogeneity (e.g., one could assume instead Križ and Spector's (2017) 'truth-on-all-readings' account of homogeneity, following Krifka 1996). While this works to explain (35a), we point out that this theory would expect infelicity also in cases where we have argued there is no infelicity, such as (29) and (33). In both examples, the context entails **Some**, and assuming that the sentences have an existential reading, which is contextually tautologous, the sentences should be infelicitous on Spector's account. In contrast, if one assumes the implicature account of homogeneity together with Magri's theory of mandatory implicatures, triviality only leads to infelicity if it is the result of a strengthened meaning, so this problem does not arise.

¹¹ Guerrini and Wehbe (2024) themselves assume a version of the implicature account of homogeneity, albeit one that takes implicatures to be presuppositional (following Bassi et al. 2021) and therefore predicts HP. In section 7 we will discuss a potential direction for reconciling their account with our observations.

¹² See Doron and Wehbe (2024) for some relevant discussion of LA and PAI (though they do not discuss the ability of stress to rescue PAI violations).

- (41) I knew that Mary doesn't smoke now, but today I learned that...
- a. She {STOPPED, #stopped} smoking. (She used to be a heavy smoker!)
- (42) I knew that Mary is planning to speak in the conference about a book she wrote. But today I learned that...
- a. She will be speaking about {THE, #the} book she wrote. (She has just one!)
- (43) I already knew that Mary is on sabbatical this year. But today I realized that...
- a. She's on sabbatical {AGAIN, #again}. (She took a sabbatical before, she's so lucky!)

The generalization from these data seems to be that LA is possible only if main stress falls on the relevant presupposition trigger, and this is in line with a similar claim made in Bassi et al. (2021). Note that stress on anything else in (41)-(43) leads to infelicity. If that is so, then the inability of the trigger of HP in definite plurals to bear focal stress, the factor which we identified as relevant for explaining infelicity with homogeneity (sections 4 and 5.2), could now also be thought of as the reason why LA cannot save infelicity with homogeneity in most cases. But even so, we are still short of explaining what's going on in the felicitous examples (29) and (33a), because these examples do not pattern accordingly: in (29) there is polarity focus, and in (33a) stress falls on some nominal within the definite plural; presumably, in neither case could the stressed element be considered the HP trigger.

This is not to say that the direction we just sketched, in terms of a stress condition on LA, is hopeless against the challenge posed by (29) and (33a). Here is a broad sketch of a possible speculative direction worthy of pursuit if one continues this line of thought. It is based on combining the implicature account of homogeneity (see section 6) with the 'presuppositional implicatures' account by Bassi et al. (2021), which is incidentally precisely the account Guerrini and Wehbe (2024) assume. We assumed here that $\mathcal{E}xh$ is responsible for implicature computation, and that the inferences $\mathcal{E}xh$ computes are at-issue content. Bassi et al. (2021) however argue that these inferences are non-at-issue content, and accordingly propose instead that implicature computation is done by $\mathcal{P}ex$, an exhaustivity operator that presupposes the inferences that $\mathcal{E}xh$ is usually taken to assert. Once Bassi et al.'s $\mathcal{P}ex$ theory is combined with the implicature account of homogeneity, homogeneity is predicted as a presupposition. One may ask then in general what stress pattern allows for LA when the presupposition trigger is $\mathcal{P}ex$: It can't possibly be $\mathcal{P}ex$ itself, which is covert. The generalization seems to be, based on Bassi et al.'s own observation, that LA's stress condition is satisfied if stress falls on a focus associate of $\mathcal{P}ex$. One can then imagine that, when there are several alternative-generating items that $\mathcal{P}ex$ associates with, stress on one of them should be enough to allow for LA. If that is on the right track, one could explain the felicity of (33a) given that $\mathcal{P}ex$ associates with both *phonology* and the covert operator that is responsible for homogeneity, so stress on *phonology* suffices to allow LA. As for (29), this will presumably require assuming that $\mathcal{P}ex$ associates with both the covert operator that is responsible for homogeneity and the polarity head; whether this is reasonable is a question we will leave to another occasion.¹³

7.2 Homogeneity beyond definite plurals

Another issue we should mention before wrapping up, which may be taken to provide an argument for our focus-based account, concerns other domains where homogeneity effects have been observed.

¹³ For a view where the polarity head is a focus associate of an exhaustivity operator see Bassel (2023).

Not all of them behave uniformly with respect to infelicity with homogeneity, as WG point out. The sentences in (44) (based on example (44) in [Guerrini and Wehbe 2024](#)) replicate WG’s observation with definite plurals in other domains: Habituals ((44a); see [Agha 2023](#)), generics ((44b); see [Löbner 2000](#); [Križ 2015](#); [Agha 2023](#)), episodic universal bare plurals ((44c); see [Condoravdi 2016](#); [Chierchia 2022](#)), predicate homogeneity ((44d); see [Križ 2015](#); [Paillé 2024, 2025](#)), and conditionals ((44e); see [von Fintel 1997](#); [Herburger 2015](#); [Križ 2015](#); [Bassi and Bar-Lev 2016](#); [Agha 2023](#)).

- (44)
- a. I knew that Mary sometimes goes to the gym on Tuesdays, but today I learned that. . .
 - (i) #Mary goes to the gym on Tuesdays.
 - b. I knew that wolves sometimes hunt in packs. But today I learned that. . .
 - (i) #wolves hunt in packs.
 - c. I knew that some birds are migrating right now. But today I learned that. . .
 - (i) #Birds are migrating right now.
 - d. I knew that the wall was partly red. But today I learned something interesting. . .
 - (i) #The wall is red.
 - e. I knew that if you hit the brakes the car sometimes stops, but today I learned that. . .
 - (i) #If you hit the brakes, the car stops.

Parallel examples in other domains that show what looks like homogeneity effects, however, behave differently. Specifically, with free choice disjunction ((45a); see particularly [Goldstein 2018](#); [Bar-Lev 2018, 2021](#) for the connection between free choice and homogeneity) and with neg-raising ((45b-d); see [Gajewski 2005](#); [Staniszewski 2019](#); [Jeretič 2021](#); [Agha and Jeretič 2022](#)), there is no infelicity in parallel contexts, as long as the sentence is uttered with a particular stress pattern, as (45) demonstrates ((45a) and (45d) are based on [Guerrini and Wehbe 2024](#), exx. (42) and (47)):

- (45)
- a. I knew that Mary can eat the cake and I wasn’t sure about the candy. but today I learned that. . .
 - (i) she can eat the cake {OR, #or} the candy.
 - b. I knew that Mary is fine with meeting over zoom, but today I learned that. . .
 - (i) she {WANTS, #wants} to meet over zoom.
 - c. I knew that Mary thinks it’s possible that the earth is flat, but today I learned that. . .
 - (i) she (actually) {BELIEVES, #believes} the earth is flat.
 - d. I knew that Mary is allowed to read this book, but today I learned that. . .
 - (i) she {SHOULD, #should} read this book.

If it is desirable to maintain a unified treatment of homogeneity phenomena across domains, one may take the contrast between (44) and (45) as support for the view we proposed here (as WG mention), given that the distinction between free choice and neg-raising on the one hand and all the other domains on the other hand tracks the difference between having an overt connective or quantificational element (*or* in the case of free choice, *want/believe/should* in the case of neg-raising) and not having any such thing (all other domains), and that stress on these overt items is required. Whether this is indeed a strong argument in favor of our approach is not entirely clear to us for two reasons: First, the verdict is still out on whether homogeneity phenomena should be treated uniformly. Second, as we just discussed, PAI violations can be avoided with stress on presupposition triggers, which potentially provides an alternative explanation

for the felicity of the examples in (45) on accounts that assume HP.¹⁴

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¹⁴ This would presumably require assuming that *or*, *want*, *believe*, and *should* are either the triggers of HP in the examples in (45), or are focus associates of a presuppositional exhaustivity operator that derives HP (based on our discussion above). This may provide a way to tease apart theories of homogeneity effects in these domains: For instance, on Goldstein’s (2018) account of free choice disjunction, the trigger of HP is the modal, which makes it surprising that stress falls on *or* in (45a-i). In contrast, this is expected if disjunction is the focus associate of a presuppositional exhaustivity operator that derives HP, as in Del Pinal et al. (2024)

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